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Title:- Smart Hospital Management System using Doubly Linked list and Queue in C++

problem Statement :-
managing patient registration, maintaining medical visit record, handling waiting queues, and tracking patient discharge manually can be time-consuming and inefficient. It may lead to delays in patient processing difficulty retrieving patient information, and challanges in monitoring hospital operations.
To develop a Smart Hospital management System using C++ that simplifies patient registration, queue management and discharge tracking. The system uses a Doubly linked list to maintain patient record and support forward and backword traversal. while a FIFO Queue ensures that patient are called in their registration order.

Objective:-
1) To maintain patient record using a doubly linked list.
2) To manage waiting patients using a FIFO Queue
3) To provide patient registration and automatic record insertion.

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4) To search for patient information using unique patient ID
5) To day patient record efficiently.
6) To track patient consulation and discharge status
7) To demonstrate dynamic memory allocation and pointer manipulation in C++.

Algorithm

Step 1: Start the program.
Step 2: Initialize the FIFO Queue pointers front and rear to NULL and set the initial patient token number to 1001.
Step 3: Initialize the doubly linked list pointers head and tail to NULL
Step 4: Display the main menu with the following operations
1) Register patient
2) call next patient
3) Search patient
4) Display All patient Records
5) Display records in Reverse
6) count waiting Patients
7) Hospital Dashboard
8) Discharge patient
9) Exit.

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Step 5 :- Accept the user's choice
Step 6 :- Register patient
1) Create a new patient node dynamically.
2) Enter the patient's name, age and department.
3) Generate a unique token number and initialize the patient-status to waiting.
4) insert the patient at the end of doubly linked list by updating the prev, Record and next pointers.
5) insert at patient at the rear of the fifo Queue using the next Queue pointer.
6) Display the token number, queue position and estimated waiting time

Step 7 :- Search patient
1) Accept the patient's token number
2) Start traversing the Doubly linked list from head.
3) compare the token number of each patient with the entered token.
4) if a match is found, display the patient's details and current status.
5) if no match is found, display "patient's not found".

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Step9 :- Display All patient Records
1) check whether the doubly linked list is empty
2) if it is empty, display a suitable message.
3) otherwise, start from head
4) Display each patient token number, name, age, department, and status.
5) move to the next node using next Record until the end of the list is reached.

Step10 :- Display Record in Reverse
1) check whether the doubly linked list is empty
2) if is is not empty, start from rail
3) Display the current patients details.
4) move to the previous node using Rev PrevRecord.
5) Repeat until NULL is reached.

Step11 :- Count waiting patients
1) initialize the waiting count to zero
2) Traverse the FIFO Queue from front using next queue.
3) increment the count of each patient in the Queue
4) Display the total number of waiting Patients.

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Step 12: Display Hospital Dashboard
1) Initialize the total, waiting, served, and discharged counters to zero.
2) Traverse the doubly linked list.
3) Count all registered patient and classify them according to their status.
4) Display the total number of registered waiting, served and discharged patients
5) Display the estimated queue waiting time

Step 13: Discharge patient.
1) Accep the patient's token number
2) Search for the patient in the doubly linked list.
3) if the patient is not found, display "Patient not found"
4) if the patient is still waiting do not allow discharge.
5) otherwise discharge patient.

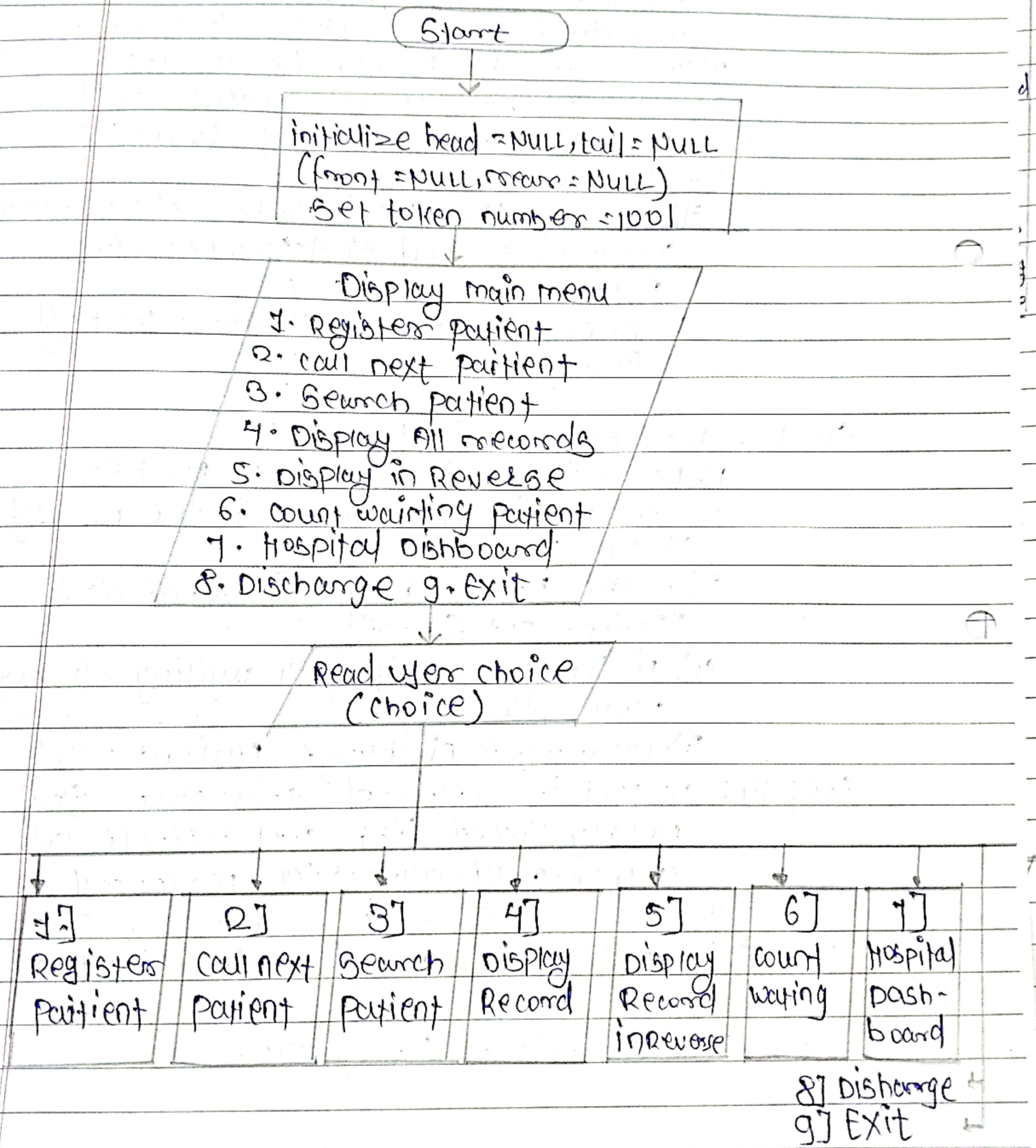
Step 14: if exit is selected traverse the Doubly linked list and release all dynamically allocated memory.

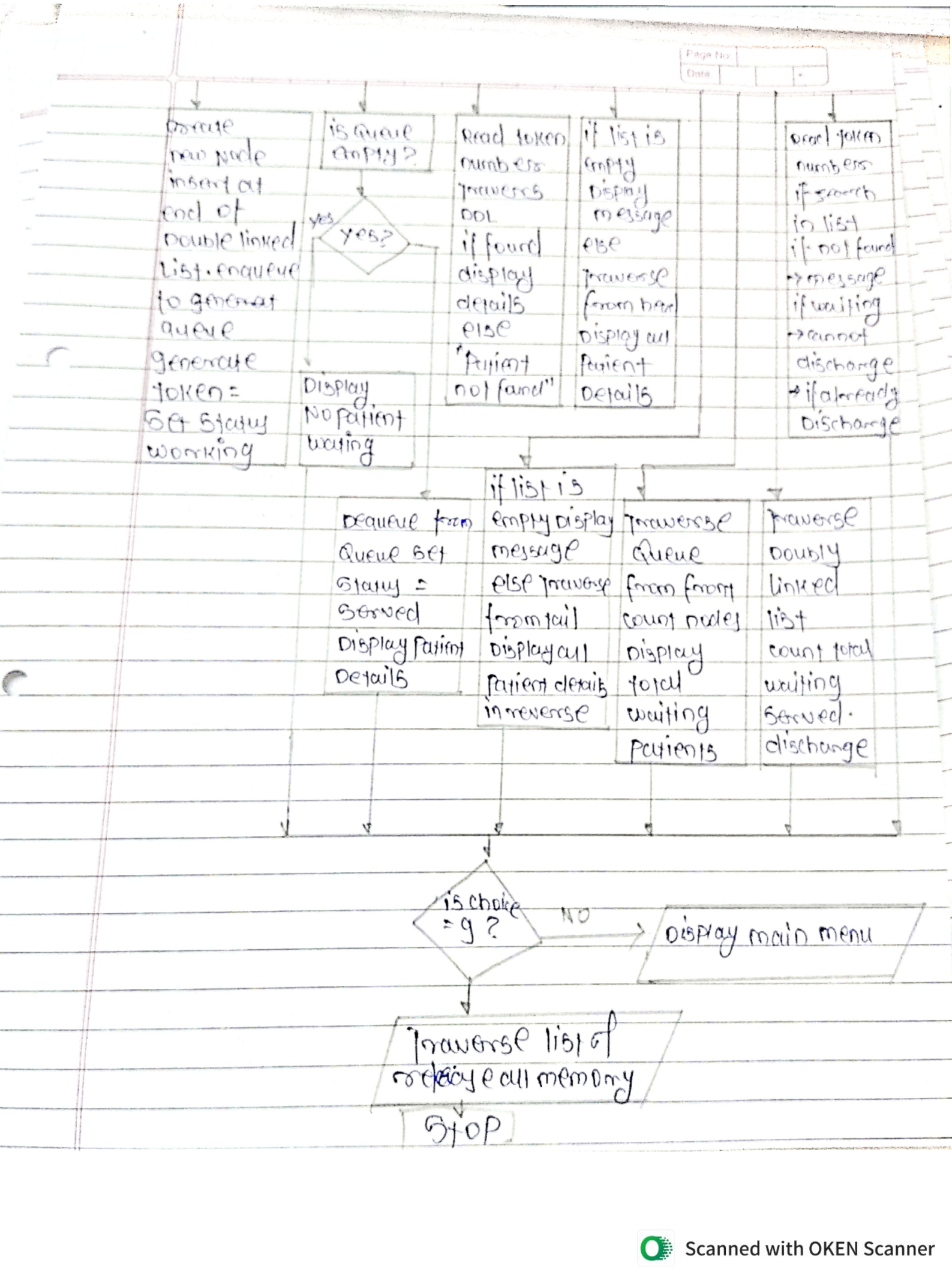
Step 15: Display the program termination message

Step 16: Stop the program.

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flowchart




create new node insert at end of double linked list. enqueue to general queue generate token = Set Status working
is Queue empty?
yes
yes?
Display No patient waiting
Read token numbers traverse DDL if found display details else "Patient not found"
if list is empty Display message else traverse from head Display all Patient Details
Read token numbers if search in list if not found → message if waiting → cannot discharge → if already Discharge
Dequeue from Queue Set Status = Served Display Patient Details
if list is empty display message else traverse from tail Display all Patient details in reverse
Traverse Queue from front count nodes display total waiting Patients
Traverse Doubly linked list count total waiting Served. discharge
is choice = 9 ?
NO
display main menu
Traverse list of release all memory
Stop